

Role of *Slc13a5* in osteoblast-mediated bone remodeling and marrow fat regulation

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Introduction

Bone remodeling is regulated by mechanical loading, a critical process for maintaining bone health, and when impaired, can lead to pathology such as osteoporosis. Citrate is a critical component of bone matrix, mineralization, and bone strength, while also playing a role in cell metabolism [1]. The sodium-dependent citrate transporter encoded by *Slc13a5* (NaCT) enables citrate uptake into osteoblasts. Because osteoblasts and adipocytes share a common mesenchymal progenitor, altered osteoblast metabolism may influence marrow adiposity. However, the role of osteoblast-specific citrate transport in regulating bone structure and marrow fat balance remains unclear. This study investigates whether osteoblast-specific deletion of *Slc13a5* alters bone remodeling and marrow adiposity. We hypothesized that impaired citrate transport in osteoblasts would blunt the bone remodeling response to mechanical loading and promote adipocyte formation.

Methods

•**Animal model:** Osteoblast specific conditional knockout of *Slc13a5* (*Slc13a5* cKO) and Cre recombinase negative (Cre⁻) littermate control mice were used in this experiment.

•**In vivo microCT & Computational Registration:** *In vivo* microcomputed tomography (microCT) scans were registered using Dragonfly to provide spatial insights into active areas of bone remodeling (Figure 1).

•**Mechanical Loading:** Tibial loading was used to evaluate the bone response to mechanical loading in cKO and Cre⁻ mice. For both genotypes, the right tibia was loaded while the left served as a non-loaded control. Loading was performed 5 days/week for 2 weeks.

•**Ex vivo microCT Analysis:** MicroCT scans were also analyzed using Scanco software to assess differences in bone phenotypic parameters between cKO and Cre⁻ mice under loaded and non-loaded conditions.

•**Marrow Adiposity Quantification:** Osmium staining was used to visualize and quantify marrow fat volume in cKO and Cre⁻ groups (Figure 2).

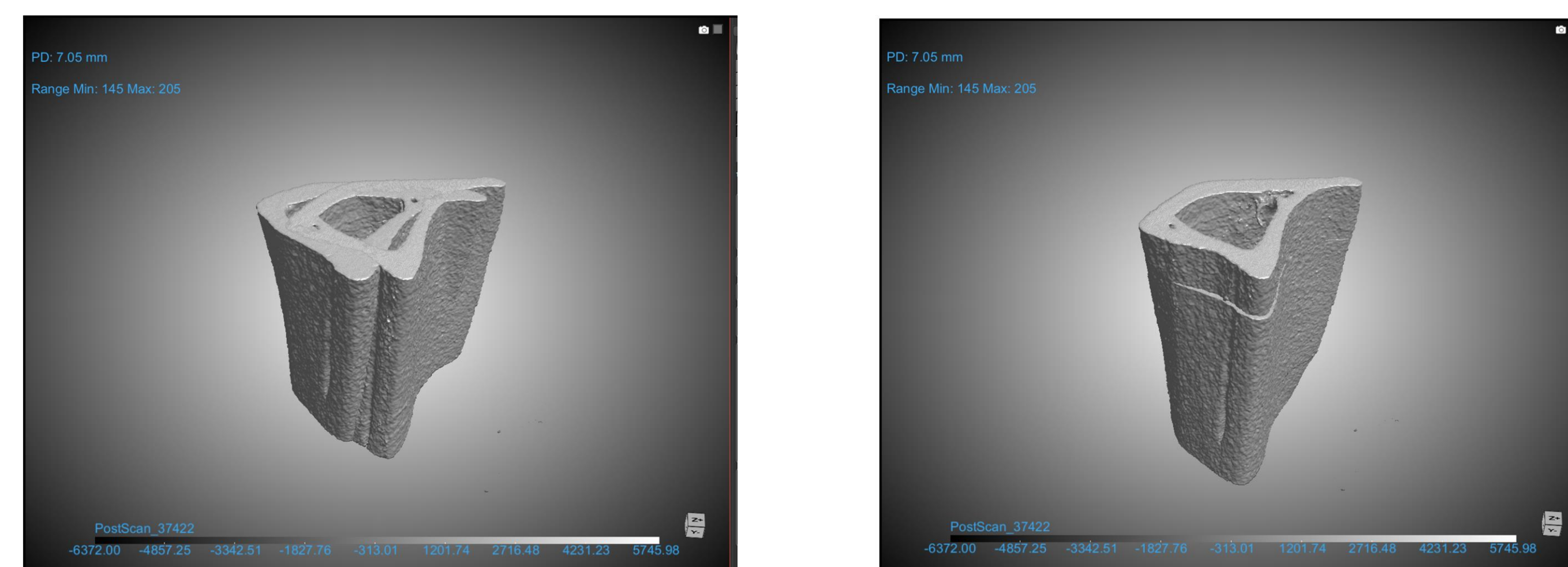


Figure 1: Representative *in vivo* microCT images before and after registration using Dragonfly software, illustrating alignment of scans to enable spatial analysis of bone remodeling.



Figure 2: Osmium staining used to visualize bone marrow adipose tissue (BMAT) within bone.

Results

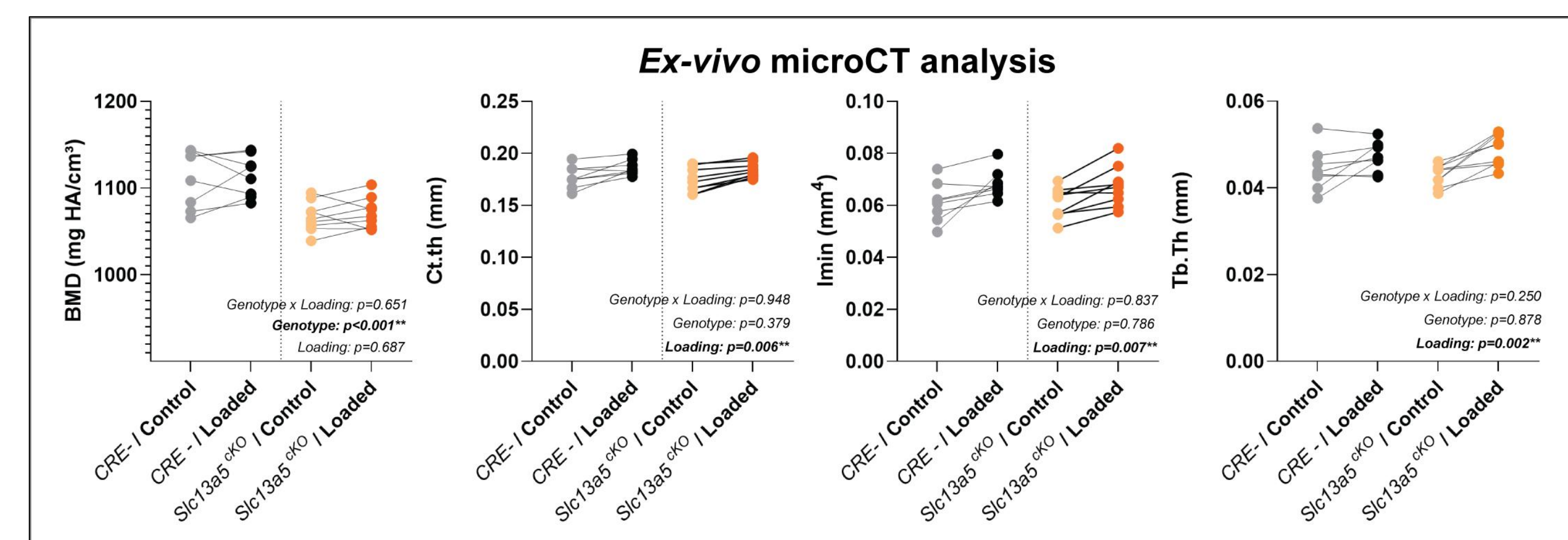


Figure 3: *Ex vivo* microCT analysis of bone parameters in Cre⁻ and *Slc13a5* cKO mice under loaded and non-loaded conditions. [2]

Mechanical loading showed significant increase in cortical thickness (Ct.Th), second moment of area (Imin), and trabecular thickness (Tb.Th) in both groups. Additionally, there was no difference in response to loading between cKO and Cre⁻ control groups. Bone mineral density (BMD) was lower in cKO mice compared to Cre⁻ groups (Figure 3).

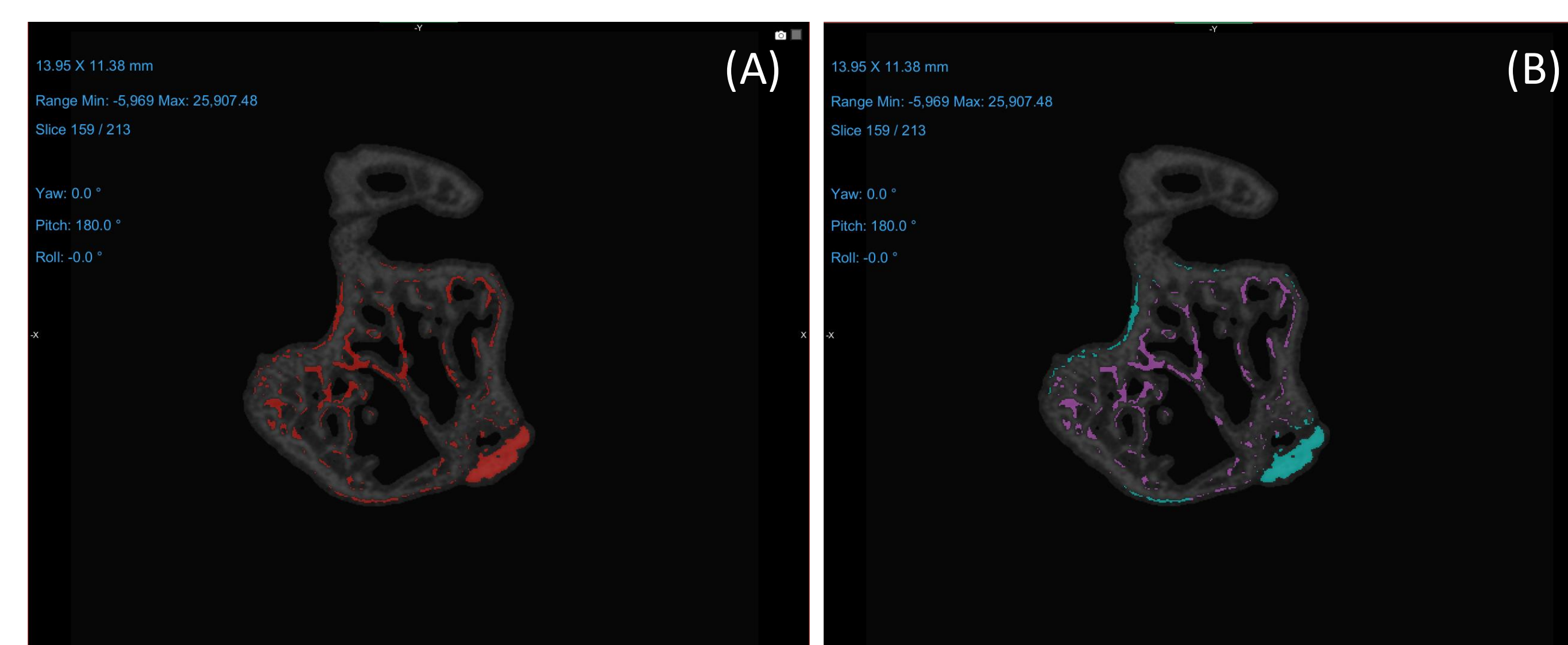


Figure 4: Bone formation in a cKO mouse. (A) Total bone formation (red overlay). (B) Spatial distribution of bone formation, with endosteal formation shown in magenta and periosteal formation shown in blue.

Bone formation was observed at both endosteal and periosteal surfaces, indicating spatially distributed remodeling in response to loading. Quantification showed a trend toward increased resorption and formation in the cKO groups compared to Cre⁻, though the difference was not significant (Figure 5).

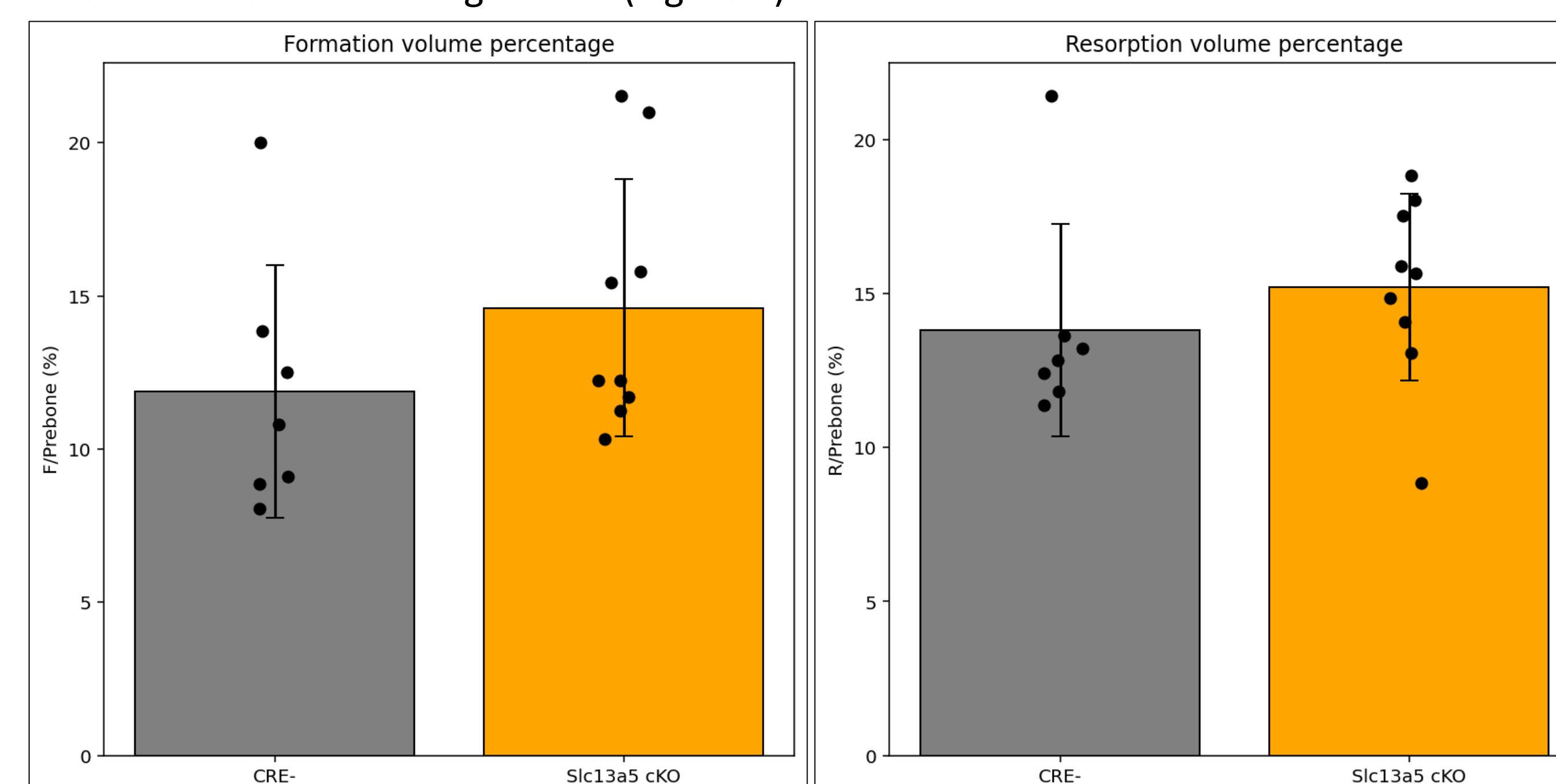


Figure 5: Formation and resorption bone volume percentages in Cre⁻ and *Slc13a5* cKO mice.

Results

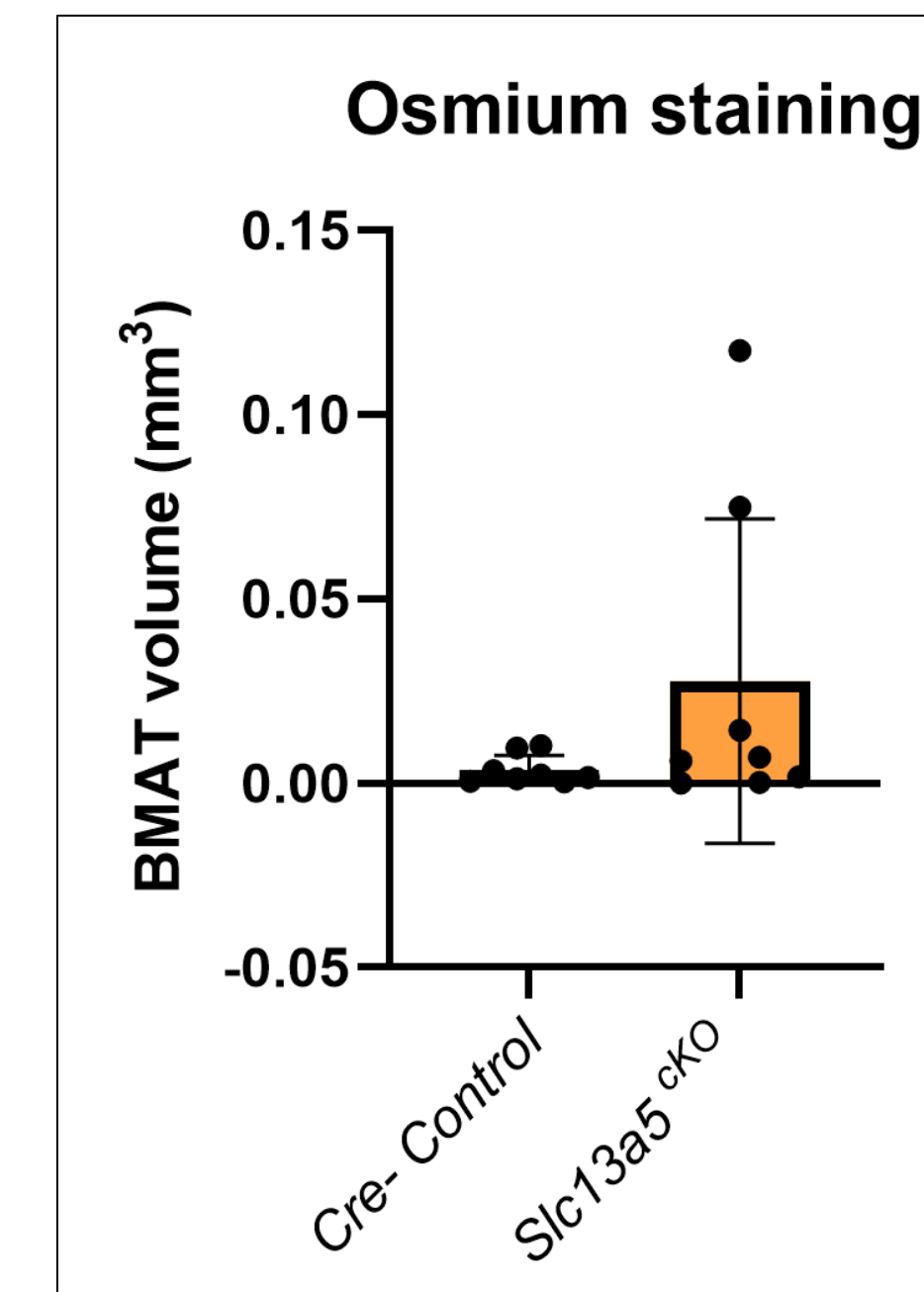


Figure 6: Bone marrow adipose tissue (BMAT) volume measured by osmium staining in Cre⁻ and *Slc13a5* cKO mice. We observed a trend toward an increase in BMAT volume in *Slc13a5* cKO compared to Cre⁻. However, the difference between groups was not significant.

Discussion and Conclusions

Deletion of the citrate transporter encoded by *Slc13a5* in osteoblasts did not impair the skeletal response to mechanical loading, as both Cre⁻ and cKO mice exhibited similar increases in cortical and trabecular bone parameters (Figure 3). Consistent with this, formation and resorption volume percentages were not significantly different between groups (Figure 5), indicating that overall bone remodeling activity is preserved despite loss of citrate transport in osteoblasts. Representative spatial mapping further demonstrated that bone formation occurred at both endosteal and periosteal surfaces, indicating that the distribution of remodeling activity is preserved despite loss of osteoblast-specific citrate transport (Figure 4).

No significant differences in marrow adiposity were observed between groups, suggesting that disruption of citrate uptake in osteoblasts does not significantly influence adipocyte formation in femurs from adult mice (Figure 6). These findings do not support our hypothesis that impaired citrate transport in osteoblasts would weaken the bone remodeling response to mechanical loading and increase adipocyte formation.

These findings suggest that osteoblast-specific citrate transport is not required for load-induced bone formation. Ongoing computational analyses will provide further insight into the spatial dynamics of bone remodeling and potential changes in structural organization (Figure 4).

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References

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